

1.13.10. Multi Position Details

This section will discuss some archetypes of switch devices with more than 2 positions and how they SHALL set attribute values and generate events matching their characteristics.

For the SwitchLatched, InitialPress, LongPress and MultiPressOngoing events, the field NewPosition SHALL be set to the value corresponding to the new position to which the switch was moved. For the ShortRelease, LongRelease and MultiPressComplete events, the field PreviousPosition SHALL be set to the value corresponding to the position of the switch just preceding the (latest) release.

1.13.10.1. Latching Switch with N stable positions ($N > 2$) with fixed sequence

With such a device, the user can move the switch from a position M to positions M-1 and M+1 (either with a wraparound between the end positions, or fixed stop at the end positions).

On each interaction with the switch device, it SHALL generate a SwitchLatched event with the NewPosition field set to the value associated with the new position.

Due to the physical constraints, such an event will have a NewPosition field which is equal to the previous NewPosition field plus or minus 1 (modulo NumPositions if the switch does not have end stops).

In a first example, a switch has 3 positions, associated with values 0, 1 and 2. In this case, wrap-around is not possible: from position 0 it can only be moved to position 1.

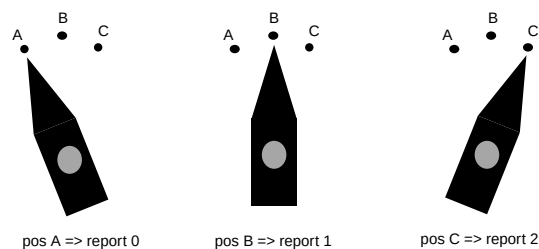


Figure 8. Rotary switch device with 3 positions

In another example, a switch has 8 positions, associated with values 0 through 7. In this case, the physics of the switch allow wraparound: from position 0 it can be moved to position 1 or to position 7.